

Experiment_2 Group2E

December 11, 2025

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[25]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
import math
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[26]: # 1. RAW DATA

time = np.array([0, 15, 30, 45, 60])    # seconds

# Digital thermometer data (°C)
cu_dt = np.array([20, 22, 30, 47, 66])
al_dt = np.array([19, 24, 38, 57, 68])
ss_dt = np.array([20, 22, 25, 33, 42])

# IR data (°F)
cu_ir_f = np.array([70.1, 73.5, 77.4, 84.2, 89.5])
al_ir_f = np.array([70.1, 72.7, 78.2, 84.2, 87.9])
ss_ir_f = np.array([70.1, 70.9, 73.5, 76.4, 75.9])

# Convert IR readings to °C
cu_ir = (cu_ir_f - 32) * 5/9
al_ir = (al_ir_f - 32) * 5/9
ss_ir = (ss_ir_f - 32) * 5/9
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[27]: # 2. LINEAR REGRESSION: HEATING RATES

def heating_rate(time, temp):
    model = LinearRegression().fit(time.reshape(-1,1), temp)
    return model.coef_[0], model.intercept_

cu_slope, cu_b = heating_rate(time, cu_dt)
al_slope, al_b = heating_rate(time, al_dt)
ss_slope, ss_b = heating_rate(time, ss_dt)

print("Heating Rates (°C/s):")
print(f"  Copper:      {cu_slope:.3f}")
print(f"  Aluminum:     {al_slope:.3f}")
print(f"  Stainless:     {ss_slope:.3f}")
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print()
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Heating Rates (°C/s):

Copper:	0.780
Aluminum:	0.873
Stainless:	0.367

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[28]: # 3. PLOT TEMPERATURE VS TIME (DIGITAL THERMOMETER) - BEST FIT LINES
plt.figure(figsize=(10,6))
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# Scatter points
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plt.scatter(time, cu_dt, label="Copper Data")
plt.scatter(time, al_dt, label="Aluminum Data")
plt.scatter(time, ss_dt, label="Stainless Steel Data")
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# Best-fit lines
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time_lin = np.linspace(0, 60, 200).reshape(-1,1)
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cu_fit = cu_b + cu_slope * time_lin
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al_fit = al_b + al_slope * time_lin
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ss_fit = ss_b + ss_slope * time_lin
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plt.plot(time_lin, cu_fit, label="Copper Best Fit")
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plt.plot(time_lin, al_fit, label="Aluminum Best Fit")
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plt.plot(time_lin, ss_fit, label="Stainless Steel Best Fit")
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plt.xlabel("Time (s)")
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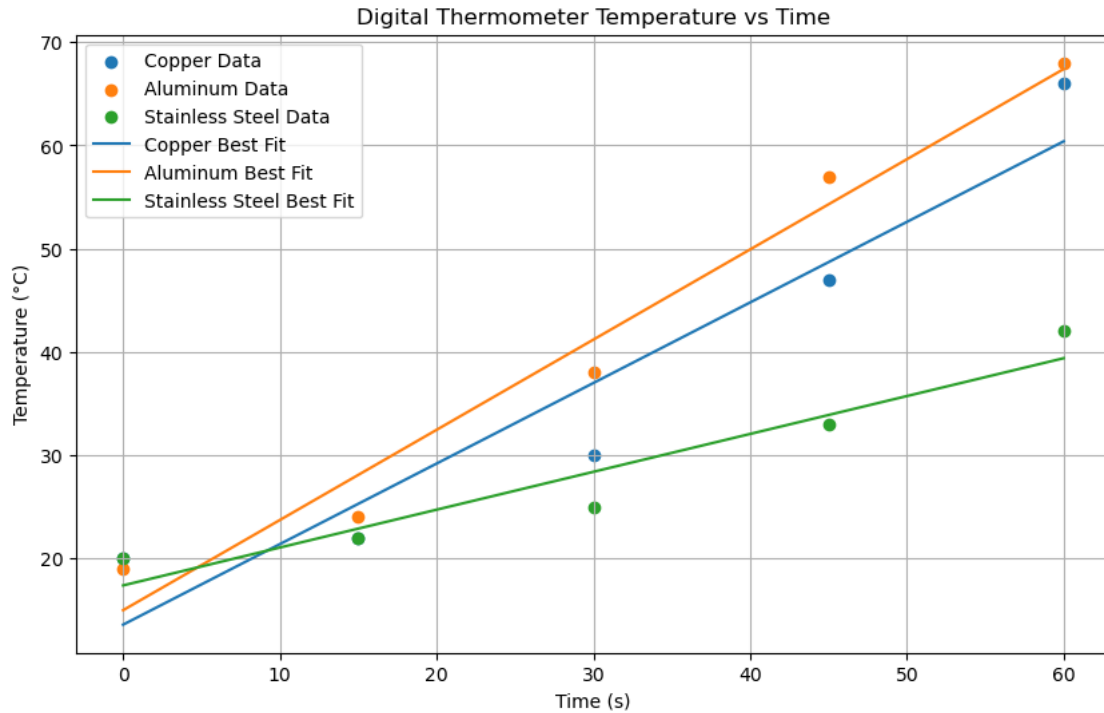
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plt.ylabel("Temperature (°C)")
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plt.title("Digital Thermometer Temperature vs Time")
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plt.legend()
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plt.grid(True)
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plt.show()
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[29]: # IR CAMERA PLOT - BEST-FIT LINES
plt.figure(figsize=(10,6))

# Scatter points
plt.scatter(time, cu_ir, label="Copper IR Data")
plt.scatter(time, al_ir, label="Aluminum IR Data")
plt.scatter(time, ss_ir, label="Stainless Steel IR Data")

# Linear fit lines
time_lin = np.linspace(0, 60, 200).reshape(-1,1)

# Fit Copper
cu_ir_slope, cu_ir_int = heating_rate(time, cu_ir)
plt.plot(time_lin, cu_ir_int + cu_ir_slope * time_lin, label="Copper IR Best_
↳Fit")

# Fit Aluminum
al_ir_slope, al_ir_int = heating_rate(time, al_ir)
plt.plot(time_lin, al_ir_int + al_ir_slope * time_lin, label="Aluminum IR Best_
↳Fit")

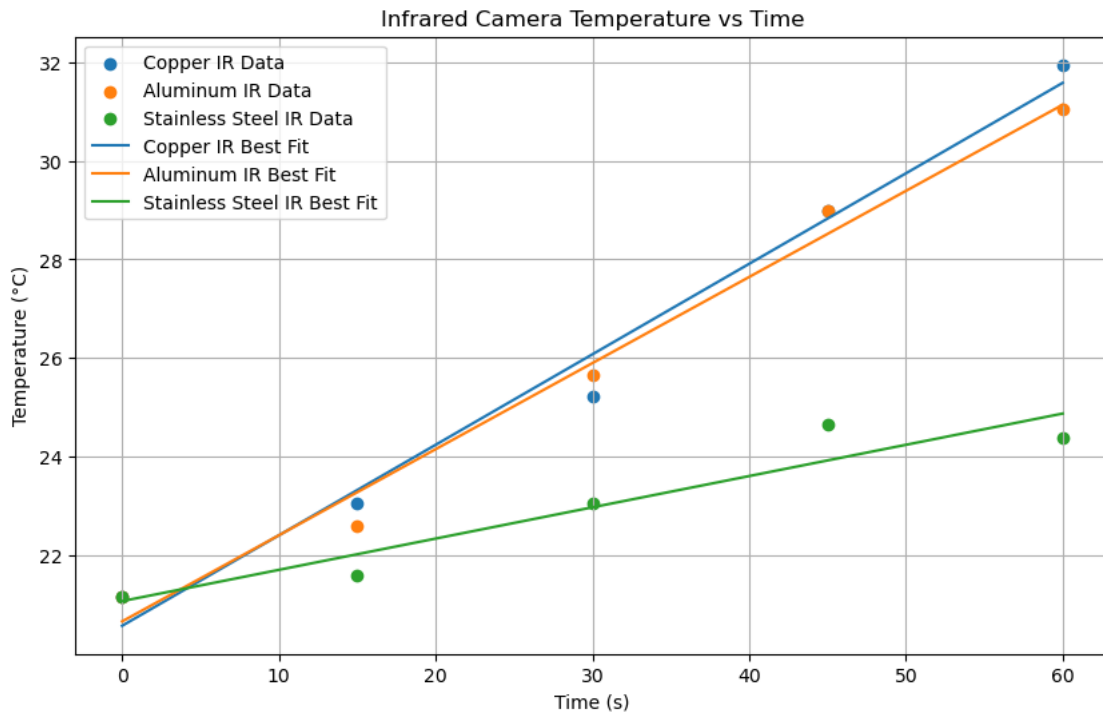
# Fit Stainless Steel
ss_ir_slope, ss_ir_int = heating_rate(time, ss_ir)
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plt.plot(time_lin, ss_ir_int + ss_ir_slope * time_lin, label="Stainless Steel IR Best Fit")

plt.xlabel("Time (s)")
plt.ylabel("Temperature (°C)")
plt.title("Infrared Camera Temperature vs Time")
plt.legend()
plt.grid(True)
plt.show()

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[30]: # 4. IR ERROR ANALYSIS
print("Average IR Measurement Error (°C):")
print(f"  Copper:    {np.mean(cu_err):.2f}")
print(f"  Aluminum:   {np.mean(al_err):.2f}")
print(f"  Steel:      {np.mean(ss_err):.2f}")

plt.figure(figsize=(10,6))

plt.scatter(time, cu_err, label="Copper IR Error")
plt.plot(time, cu_err, ':. ', label="Copper Trend")

plt.scatter(time, al_err, label="Aluminum IR Error")
plt.plot(time, al_err, ':. ', label="Aluminum Trend")

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plt.scatter(time, ss_err, label="Steel IR Error")
plt.plot(time, ss_err, ':. ', label="Steel Trend")

plt.axhline(0, color='black', linewidth=0.8)

plt.xlabel("Time (s)")
plt.ylabel("IR Error (°C)")
plt.title("Infrared Error Relative to Digital Thermometer")
plt.legend()
plt.grid(True)
plt.show()

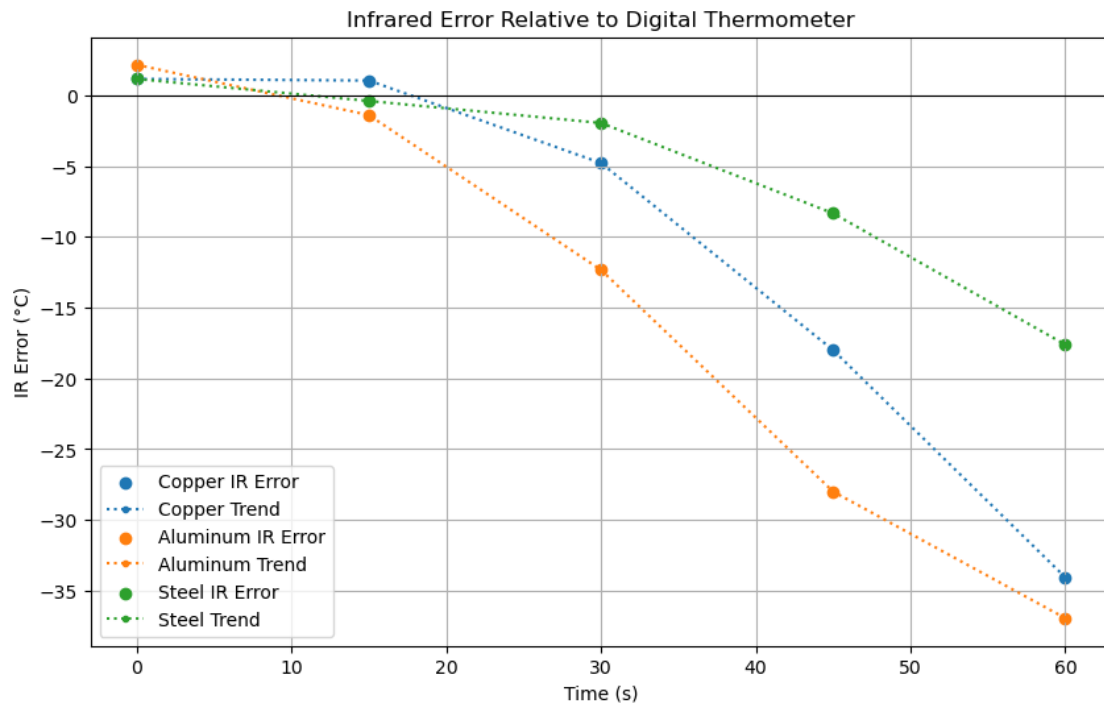
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Average IR Measurement Error (°C):

Copper: -10.92

Aluminum: -15.30

Steel: -5.42



[31]: # 5. NORMALIZED HEATING RATE (Rate / Cross-sectional Area)

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# Diameters (m)
d_cu = 3.34e-3
d_al = 2.95e-3
d_ss = 2.95e-3

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A_cu = math.pi * (d_cu/2)**2
A_al = math.pi * (d_al/2)**2
A_ss = math.pi * (d_ss/2)**2

print("Cross-sectional Areas (m²):")
print(f"  Copper:      {A_cu:.8e}")
print(f"  Aluminum:     {A_al:.8e}")
print(f"  Stainless:    {A_ss:.8e}\n")

print("Normalized Heating Rate (°C/s per m²):")
print(f"  Copper:      {cu_slope/A_cu:.2e}")
print(f"  Aluminum:     {al_slope/A_al:.2e}")
print(f"  Stainless:    {ss_slope/A_ss:.2e}")

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Cross-sectional Areas (m²):
  Copper:      8.76158775e-06
  Aluminum:     6.83492752e-06
  Stainless:    6.83492752e-06

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Normalized Heating Rate (°C/s per m²):
  Copper:      8.90e+04
  Aluminum:     1.28e+05
  Stainless:    5.36e+04

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